3. Which of the following sets are equivalent or unequal or equal sets?

(i) A = The set of vowels in the English alphabets.

B = The set of all letters in the word "VOWEL"

(ii) $C = \{2,3,4,5\}$ $D = \{ x : x \in \mathbb{W}, 1 < x < 5\}$

(iii) X = { x : x is a letter in the word "LIFE"} $Y = \{ F, I, L, E\}$

(iv) $G = \{ x : x$ is a prime number and $3 < x < 23\}$ $H = \{ x : x$ is a divisor of $18\}$

4. Identify the following sets as null set or singleton set.

(i) $A = \{x : x \in \mathbb{N}, 1 < x < 2\}$

(ii) B = The set of all even natural numbers which are not divisible by 2

(iii) $C = \{0\}$.

(iv) D = The set of all triangles having four sides.

5. State which pairs of sets are disjoint or overlapping?

(i) $A = \{f, i, a, s\}$ and $B = \{a, n, f, h, s\}$

(ii) $C = \{x : x$ is a prime number, $x > 2\}$ and $D = \{x : x$ is an even prime number$\}$

(iii) $E = \{x : x$ is a factor of $24\}$ and $F = \{x : x$ is a multiple of 3, $x < 30\}$

6. If S = {square, rectangle, circle, rhombus, triangle}, list the elements of the following subset of S.

(i) The set of shapes which have 4 equal sides.

(ii) The set of shapes which have radius.

(iii) The set of shapes in which the sum of all interior angles is 180°.

(iv) The set of shapes which have 5 sides.

7. If $A = \{a, \{a, b\}\}$, write all the subsets of A.

8. Write down the power set of the following sets:

(i) $A = \{a, b\}$ (ii) $B = \{1, 2, 3\}$ (iii) $D = \{p, q, r, s\}$ (iv) $E = \varnothing$

9. Find the number of subsets and the number of proper subsets of the following sets.

(i) W = {red, blue, yellow} (ii) $X = \{ x^2 : x \in \mathbb{N}, x^2 \leq 100\}$.

10. (i) If $n(A) = 4$, find $n[P(A)]$. (ii) If $n(A) = 0$, find $n[P(A)]$.

(iii) If $n[P(A)] = 256$, find $n(A)$.

1.5 Set Operations

We started with numbers and very soon we learned arithmetical operations on them. In algebra we learnt expressions and soon started adding and multiplying them as well, writing (x^2+2),$(x-3)$ etc. Now that we know sets, the natural question is, what can we do with sets, what are natural operations on them ?

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